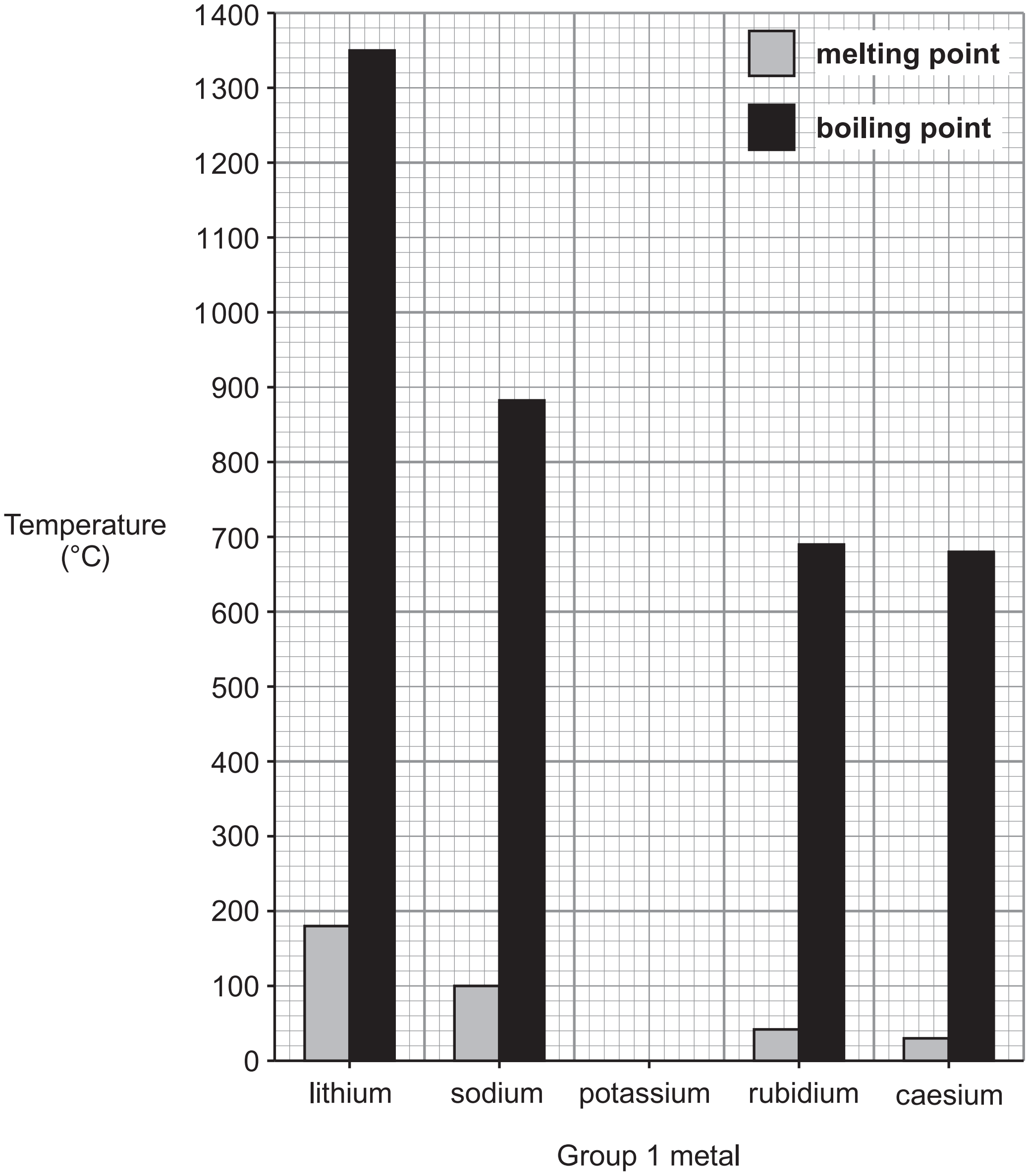
Examiner only

3. *(a)* The chart below shows the melting points and boiling points of some Group 1 metals. The data for potassium is missing.



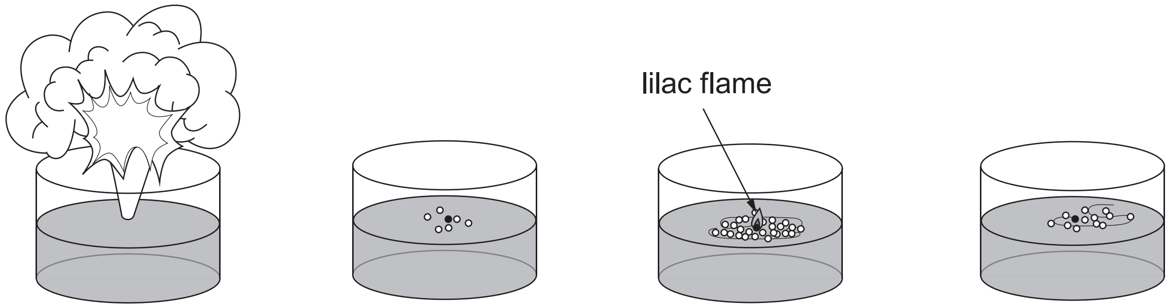
Tick (✓) the box that shows the melting point and boiling point values for potassium which fit the trends shown in the chart. [1]

melting point = 40 °C
boiling point = 600 °C ☐

melting point = 60 °C
boiling point = 780 °C ☐

melting point = 20 °C
boiling point = 600 °C ☐

(b) The diagrams show lithium, sodium, potassium and rubidium, **but not necessarily in that order**, reacting separately with cold water.



A	B	C	D
Violent reaction with sparks spitting out of the trough	Gentle fizzing	Metal ball melts, moves around on the surface and burns	Metal ball fizzes and moves around on the surface

(i) Use the information in the diagrams to give the **letter** which represents: [2]

lithium ....

sodium ....

potassium ....

rubidium ....

(ii) Caesium lies below rubidium in Group 1. Suggest how its reaction with water would be different from those above. [1]

....

(iii) Describe **one** safety precaution taken when adding a Group 1 metal to water. [1]

....

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(c) Group 1 metals react with oxygen to form metal oxides.

Sodium oxide contains the ions Na^+ and O^{2-}.

Underline the correct formula of sodium oxide. [1]

2NaO **NaO_2** **Na_2O** **Na^2O**

(d) Group 1 metals react with chlorine, Cl_2, to form metal chlorides.

Give the **letter** for the balanced symbol equation for the reaction between sodium and chlorine. [1]

A $Na + Cl \longrightarrow NaCl$

B $2Na + 2Cl \longrightarrow 2NaCl$

C $2Na + Cl_2 \longrightarrow 2NaCl$

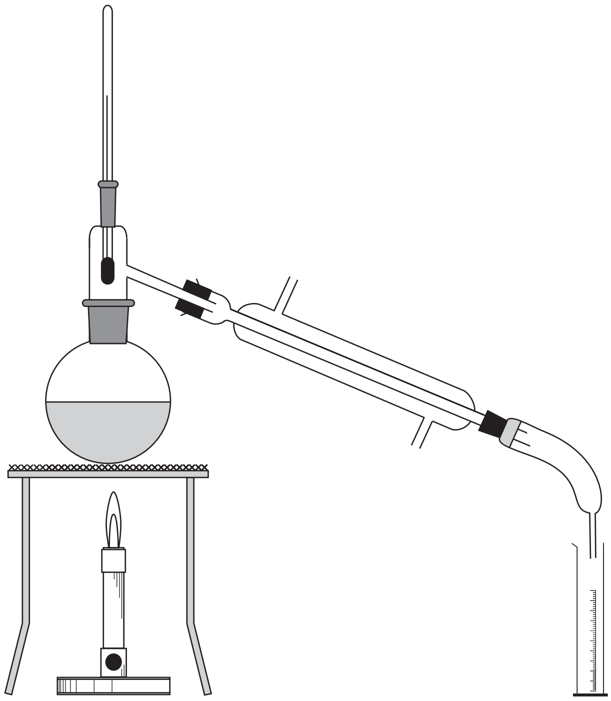
D $Na + Cl_2 \longrightarrow NaCl_2$

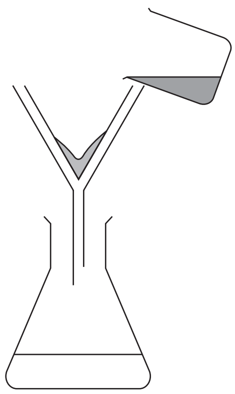
Letter ....

7

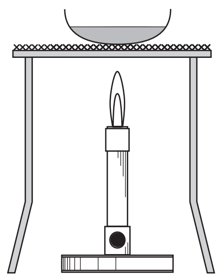
3. *(a)* The diagrams below show three methods of separating mixtures.

Method **A**





Method **B**



Method **C**

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(i) A student was given a solution of sodium chloride. State which method, **A**, **B** or **C**, he could use in order to obtain a sample of pure water from the solution. Explain how the method works. [3]

Method ....

Explanation ....

....

....

(ii) 500 g of solution was found to contain 43 g of sodium chloride. Calculate the percentage of sodium chloride in the solution. [2]

Percentage = ... %

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(b) Group 7 ions, chloride, bromide and iodide, can be identified using silver nitrate solution.

Complete the following table to show the colours of the precipitates produced by these ions. [2]

Group 7 ion	Colour of precipitate
chloride	white
bromide	....
iodide	....

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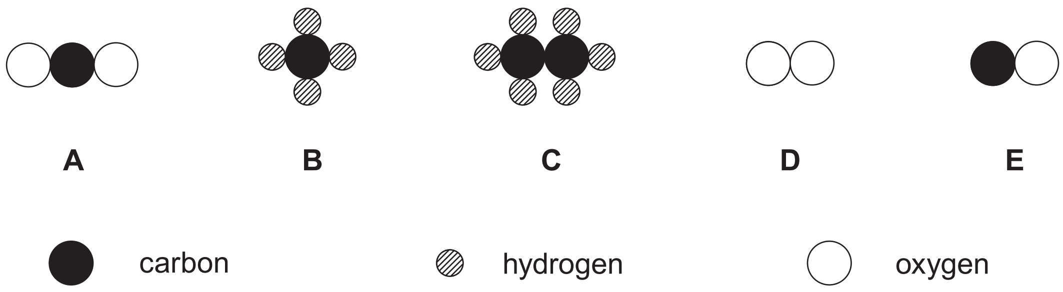
3. *(a)* The diagrams represent five substances, **A**-**E**.



(i) State which diagram, **A**-**E**, represents an element. Give a reason for your answer. [2]

Element ....

Reason ....

....

(ii) State which diagram, **A**-**E**, represents carbon dioxide, CO_2. [1]

....

(iii) Give the formula of the substance represented by diagram **C**. [1]

....

(b) When copper(II) sulfate solution reacts with sodium hydroxide, it forms copper(II) hydroxide, $Cu(OH)_2$, and sodium sulfate, Na_2SO_4.

(i) Write a **word** equation for the reaction taking place. [1]

....

(ii) Give the number of atoms of sulfur found in the formula Na_2SO_4. [1]

....

(iii) Give the **total** number of atoms in the formula $Cu(OH)_2$. [1]

....

(c) Iron(III) sulfate reacts with sodium hydroxide to produce iron(III) hydroxide.

Write the formula of iron(III) hydroxide. [1]

....

8

 (3410U10-1)

Examiner only

3. Chlorine, bromine and iodine are elements in Group 7 of the Periodic Table.

(a) State the number of electrons in the outer shell of a chlorine atom. [1]

....

(b) Underline the correct formula of chlorine gas. [1]

Cl_2 **Cl^2** **Cl** **2Cl**

(c) A teacher demonstrated the reactivity of Group 7 elements by placing heated iron wool in gas jars containing vapours of the elements.

The class recorded the results shown in the table.

Group 7 element	Observation with heated iron wool
chlorine	glowed very brightly
bromine	
iodine	glowed faintly

(i) Put a tick (✓) in the box to show the observation when heated iron wool was placed in bromine vapour. [1]

glowed less brightly than iodine ☐

glowed less brightly than chlorine ☐

glowed more brightly than chlorine ☐

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(ii) The compound produced in the reaction between iron wool and bromine contains the ions Fe^{3+} and Br^{-}.

I. Give the formula of this compound. [1]

....

II. State the name of this compound. [1]

....

(d) Tick (✓) the box next to a common use of iodine. [1]

to disinfect skin before surgery	☐
to make coloured fireworks	☐
to sterilise swimming pools	☐
to fill party balloons	☐

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6

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3. (a) (i) The diagram below shows the position of the Earth's continents today.

In 1912 Alfred Wegener suggested that all the continents must once have been joined together as one big land mass.

Diagrams **A**, **B** and **C** show the position of the Earth's continents 50 million, 100 million and 150 million years ago, but not necessarily in that order.

A

B

C

Give the letter, **A**, **B** or **C**, of the diagram which shows the position of the Earth's continents 150 million years ago. [1]

Letter ....